

DS 207: Mathematics for Data Science

Mini-Project Guidelines

PMDSC · UP Cebu · AY 2026-2027

1. Overview

The mini-project challenges student groups to apply the statistical, probabilistic, and linear-algebra concepts covered this semester to a real dataset or problem. The goal is to demonstrate not just technical correctness, but the ability to derive meaningful, business-relevant insights through proper application of course methods.

2. Group Composition

- Each mini-project shall be completed in learning teams of 2 members.
- Teams are encouraged to leverage the diverse professional backgrounds of their members when selecting a dataset or problem domain.
- Multidisciplinary teams are especially encouraged, as this mirrors real-world data science practice.

3. Scope and Topics

The project must apply concepts from at least two (2) of the three course blocks below. Projects that meaningfully integrate all three are strongly encouraged, as this best reflects the interconnected nature of real data analysis.

- Block 1: Data Foundations & EDA: descriptive statistics, distribution shape, data cleaning and treatment, correlation analysis and multicollinearity
- Block 2: Probability & Inference: probability foundations and Bayes' theorem, random variables and distributions, sampling distributions and the CLT, estimation and confidence intervals, hypothesis testing
- Block 3: Resampling & Linear Algebra: bootstrap and permutation methods, linear systems, eigenvalues and eigenvectors

4. Data Requirements

Groups must identify a dataset that serves as the foundation of their analysis. Two options are allowed:

- Field/Work-Related Data: data sourced from the professional domain of one or more team members (e.g., healthcare records, logistics data, financial transactions, retail or banking data). Groups choosing this route are expected to bring domain expertise that enriches the discussion and interpretation of results.
- Publicly Available Data: data sourced from open repositories (e.g., Kaggle, UCI ML Repository, government open data portals, PSE/company disclosures, World Bank, etc.). Groups choosing this route must demonstrate sufficient knowledge of the subject area to produce relatable, well-grounded insights and discussions.

Regardless of source, the chosen dataset must be sufficiently rich and relevant to support meaningful application of course concepts across at least two blocks.

5. Project Milestones

Milestone	Date	Details
Proposal Discussion	Oct 28, 2026	Groups present their working idea: candidate dataset, objectives, and which course blocks/topics they intend to apply. Informal, non-binding feedback given.
Consultation Session	Oct 28 – Nov 28, 2026	~0.5 to 1.5 hours reserved for one-on-one group consultations: progress check, methodology questions, preliminary results.
Submission Deadline	Dec 7, 2026, 11:59 PM	Technical Report (.ipynb) and Summary Poster both due.

Milestone	Date	Details
Final Presentation & Q&A	Week 16 (Dec 8–11) — specific day/time TBD, separate from the written final exam	Poster-led presentation; notebook available for reference during Q&A.

6. Deliverables

6.1 Technical Report (.ipynb file)

The primary deliverable is a Jupyter Notebook that integrates narrative and code into a single, cohesive document, combining properly organized Markdown cells for explanation with well-documented code cells for analysis and implementation. The notebook must follow this structure:

- Executive Summary: a concise overview of the project, key methods used, and major findings
- Introduction: background and context of the problem; motivation for the study
- Business Value: articulation of the practical value and real-world implications of the analysis
- Data Description: source, structure, variables, and any preprocessing steps
- Exploratory Data Analysis: summary statistics, distribution checks, correlation analysis, and initial observations
- Methodology: detailed explanation of the course concepts applied, including mathematical formulation and implementation, followed by the computational implementation
- Results and Discussion: presentation and interpretation of results; connection to objectives and domain context
- Conclusion and Recommendation: summary of findings, limitations, and actionable recommendations
- References: all datasets, papers, libraries, and external resources cited properly

6.2 Summary Poster (Infographic)

Each group must produce a visually engaging, infographic-style poster summarizing the key highlights of their project. Make sure it is clean, professional, and accessible to a general data science audience without requiring deep technical background to appreciate. The poster should highlight:

- The problem and its real-world relevance
- The dataset and key insights from the EDA
- The course concepts applied and how they were used
- Major results and takeaways
- Key recommendations

6.3 Use of Materials During Presentation

The poster serves as the primary visual aid during the presentation proper, guiding the flow of the team's discussion. Both the poster and the notebook are available for reference during the Q&A portion, allowing deeper examination of methodology and code.

Specific submission format requirements will be announced separately.

7. Grading Rubric

Full rubric with level descriptors follows on the next page.

Criterion (Weight)	Needs Improvement	Fair	Average	Very Good	Excellent
Practicality of Solution(25%)	Solution is impractical or irrelevant; no meaningful real-world connection.	Solution is loosely connected to the problem; applicability is superficial.	Solution is reasonable but generic; domain insight is limited.	Solution is feasible and relevant with clear domain insight; minor gaps.	Solution is highly practical, directly addresses the problem, and shows strong domain expertise.
Proper Use of Course Concepts(25%)	Concepts are absent, misidentified, or fundamentally misapplied.	Some concepts are identified but applied incorrectly or without clear purpose.	Concepts from at least two blocks are applied correctly; justification is adequate but basic.	Concepts from at least two blocks are correctly and purposefully applied; choice of method is well-justified.	Concepts spanning all three course blocks are applied with depth and precision; integration is seamless and meaningfully enhances the analysis.
Clarity of Presentation(20%)	Presentation is unclear, disorganized, or very difficult to follow.	Presentation has some structure but is hard to follow; visuals are inadequate.	Presentation is reasonably clear; flow and visuals are acceptable but unpolished.	Presentation is clear and well-organized; visuals support the narrative effectively.	Presentation is exceptionally clear, compelling, and professional; visuals significantly enhance understanding.
Q&A Responses - Individual(20%)	Members cannot answer questions or provide incorrect responses consistently.	Some members respond but answers are vague, incomplete, or mostly rely on teammates.	Most members can answer with adequate accuracy; some lack depth or confidence.	Members respond accurately and with confidence; responses show good understanding.	All members respond with depth, precision, and confidence; responses reflect genuine ownership of the work.
Coding Practices & Documentation(10%)	Code is disorganized, undocumented, or does not run. Reproducibility is absent.	Code runs but lacks documentation; structure is inconsistent and hard to follow.	Code is functional and partially documented; structure is acceptable but could be cleaner.	Code is well-structured and reasonably documented; logic is mostly clear.	Code is clean, thoroughly documented, and immediately readable. Comments and naming conventions are exemplary.

8. General Reminders

- Academic integrity is expected at all times. All code, analysis, and written content must be the team's own work. Proper citations must be given for datasets, references, and external code used.
- All team members are expected to actively contribute. Individual Q&A scoring reflects personal engagement and accountability.
- Teams are encouraged to choose problems that are genuinely interesting or meaningful to them. The best projects are those where the team has real curiosity about the outcome.
- Use of AI tools for coding assistance is permitted but must be disclosed and understood. Members must be able to explain and defend any code submitted, per the course's Generative AI policy.